

GE Healthcare

Discovery ST/Lightspeed(7.X) VCT& Pro³² 3 Phase UPS Installation

OPERATING DOCUMENTATION



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IMPORTANT PRECAUTIONS**Language****ПРЕДУПРЕЖДЕНИЕ**

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.
- V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKladU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHO PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.

ADVARSEL

(DA)

- DENNE SERVICE MANUAL FINDES KUN PÅ ENGELSK.
- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FØRSØG IKKE AT SERVICE UDSTYRET MEDMINDRE DENNE SERVICE MANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING

(NL)

- DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.
- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.
- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.

WARNING

(EN)

- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR, OR PATIENT FROM ELECTRIC SHOCK, OR FROM MECHANICAL OR OTHER HAZARDS.

HOIATUS

(ET)

- KÄESOLEV TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.
- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÖLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.
- KÄESOLEVA HOIATUSE EIRAMINE VÕIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.

VAROITUS

(FI)

- TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.
- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

ATTENTION

(FR)

- CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

(DE)

- DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.
- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

FIGYELMEZTETÉS

(HU)

- EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHEŐ EL.
- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÚTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

AÐVÖRUN

(IS)

- ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU.
- EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLD A VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLÞJÓNUSTU.
- REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓK HEFUR VERIÐ SKOÐUÐ OG SKILIN.
- BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.

AVVERTENZA

(IT)

- IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다.
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.
- 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRĪDINĀJUMS

(LV)

- ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGLŪ VALODĀ.
- JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGLŪ, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.
- NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.
- ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.

ĮSPĖJIMAS

(LT)

- ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŲ KALBA.
- JEI KLIENTO PASLAUGŲ TIEKĒJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŲ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĒ.
- NEMĒGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.
- JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO,
- MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.

ADVARSEL

(NO)

- DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.
- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

OSTRZEŻENIE

(PL)

- NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIE W JĘZYKU ANGIELSKIM.
- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.

ATENÇÃO

(PT-BR)

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENDE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENÇÃO

(PT-PT)

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENDE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE COLOCAR EM PERIGO A SEGURANÇA DO TÉCNICO, DO OPERADOR OU DO PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENȚIE

(RO)

- ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.
- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIEȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ȘI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPANATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

ОСТОРОЖНО!

(RU)

- ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.
- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСОБЛЮДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

UPOZORENJE

(SR)

- OVO SERVISNO UPUTSTVO JE DOSTUPNO SAMO NA ENGLJESKOM JEZIKU.
- AKO KLIJENTOV SERVISER ZAHTEVA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN DA OBEZBEDI PREVODILAČKE USLUGE.
- NE POKUŠAVAJTE DA OPRAVITE UREĐAJ AKO NISTE PROČITALI I RAZUMELI OVO SERVISNO UPUTSTVO.
- ZANEMARIVANJE OVOG UPOZORENJA MOŽE DOVESTI DO POVREĐIVANJA SERVISERA, RUKOVAOCA ILI PACIJENTA USLED STRUJNOG UDARA ILI MEHANIČKIH I DRUGIH OPASNOSTI.

UPOZORNENIE

(SK)

- TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.
- AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
- NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU.
- ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

ATENCION

(ES)

- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.
- SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION
- NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

VARNING

(SV)

- DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.
- OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.
- FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.
- OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.

DİKKAT

(TR)

- BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.
- EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.
- SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.
- BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a General Electric representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

IMPORTANT...RADIOACTIVE MATERIAL HANDLING

Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment.

GE Healthcare Services is required to notify the applicable U.S. state agency PRIOR to any source service event involving pin source handling. See NUC/PET Radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GE Healthcare iTrak Process to report all omissions, errors, and defects in this publication.

REVISION HISTORY

Rev	Date	Reason for change
1	July 12, 2005	Initial Release.
2	May 9, 2006	Updated to include model 9330G-14B UPS.
3	Aug. 21, 2008	Updated for Itrak1 – 13170992.

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Chapter 1

Common Instructions and Requirements

Section 1 - Scope

This installation specification is a control document for the installation of an Uninterruptible Power System (UPS) unit to be used with a General Electric Healthcare (GEHC) Computerized Tomography or PET/CT diagnostic imaging system. This document supports CT and PET/CT Systems, which use the following Power Distribution Units.

NGPDU models:

- 2326492
- 2326492-2
- 2326492-3
- 2326492-4
- 2326492-5

It also includes the instructions for interconnection with the system A1 Main Disconnect & UPS Control Panel.

Section 2 - Overview

This document was designed to guide installation of a 14kVA UPS into DST or LightSpeed 7.x (VCT, Pro³² & DVCT) systems using an NGPDU.

The UPS powers critical system computer and control functions. It is intended to bridge short power outages and provide time for crossover from normal mains power to emergency power. It is sized to provide at least 10 minutes backup with fully charged batteries.

It is recommended that if a power outage exceeds 5 minutes with no mains or emergency power, the patient should be removed from the table and an orderly shutdown of the system should be initiated.

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Main Disconnect Panel. Use of one of the following GE panels, or equal, specifically designed for use with this kit, is recommended:

- E4502AB / P5051RJ (90A)
- E4502AC / P5051RK (110A)
- E4502AE (125A)
- E4502AF (150A)

Refer to your system Pre-Installation manual for proper selection.

2.1 Tools Required

- FE Tool Kit
- LOTO
- Fluke 87 Multi-meter or equal
- Wire Stripper
- Labels

2.2 Estimated Manpower & Time Requirements

- 1 FE & 1 Electrician
- 3 hours

Section 3 - Safety Requirements

General Warning Considerations

The only user operations permitted with the UPS are:

- Starting up and shutting down of the UPS.
- Operating the user interface in the UPS.
- Connecting data interface cables.
- Monitoring the UPS.

Note: It is strongly recommended to have the "Installation and Operation Manual" for the UPS available. These operations must be performed according to the instructions in this document and the "Installation and Operation Manual" for the UPS being installed.

This document covers installation of the POWERWARE **9330G-14B** UPS only.

DANGER

THE UPS CONTAINS LETHAL VOLTAGES. SERVICE AND REPAIRS SHOULD BE PERFORMED BY AUTHORIZED PERSONNEL ONLY. THERE ARE NO USER SERVICEABLE PARTS INSIDE THE UPS.

Section 4 - Preliminary Instructions

4.1 Electrician

An electrician will be required to install a 3/4" conduit between the A1 Disconnect and the computer floor or to the general area of the UPS. The conduit should be terminated in a small wall box. This will be used for the UPS Control cable connecting the UPS and A1 Main Disconnect Panel.

4.2 Electrical Requirements

The UPS receives power from the NGPDU and returns power to the NGPDU. No other power connections are required. Low voltage and 120Vac SEO Control and interlock signals are passed between the UPS and A1 Panel.

Chapter 2

Powerware 9330G-14B UPS Installation

Perform the following installations and connections at the UPS.

Section 1 - UPS Mechanical Preparation

1.1 Site Preparation Verification

Verify the site has been prepared for the UPS installation. Locate the Powerware “Installation and Operation Manual” supplied with the UPS kit and refer to Section 1, Chapter 2 (rev E) for additional information.

1.2 Inspection

The UPS is shipped bolted to a wooden pallet as shown below and protected with an outer protective wrap. Visually verify there is no evidence of shipping damage to the packaging or UPS. Remove the outer protective wrap and perform the steps in the following section to unload the UPS from the pallet.



Figure 1 - UPS on Shipping Pallet

1.3 Unloading the UPS from the Pallet

Note: The following information has been extracted from the Powerware “Installation and Operation Manual” provided with the UPS. Refer to Section 3.2 of that document, for additional information if necessary.

CAUTION



- The UPS cabinet is heavy. Shipping weight is approximately 388kg (856lb). If unpacking instructions are not closely followed, the cabinet may tip and cause serious injury.
- Do not install a damaged cabinet. Report any damage to the carrier and contact your Eaton service representative immediately.
- Do not tilt the UPS cabinet more than 10° from vertical or the cabinet may tip over.
- Do NOT use the maintenance bypass switch housing as a handle while removing the cabinet from the pallet.

To remove the pallet:

1. If not already moved, use a forklift or pallet jack to move the cabinet to the installation site, or as close as possible, before unloading from the pallet. Insert the forklift or pallet jack forks between the skids on the bottom of the unit.
2. Remove the left front solid panel and the right front vented panel from the UPS cabinet (see Figure below). Front panels are secured with quarter-turn screw fasteners and magnetic latches. To remove the panels, unscrew the fasteners counterclockwise and pull the panels straight forward to disengage the magnetic latches.
3. If the leveling feet are not fully retracted, turn the leveling feet until they are retracted.
4. Remove the three bolts securing the rear shipping bracket to the cabinet and the three bolts securing the bracket to the pallet. Remove the rear shipping bracket. If installing the cabinet permanently, retain the hardware for later use; otherwise, discard or recycle in a responsible manner.
5. Remove the three bolts securing the front shipping bracket to the pallet. DO NOT remove the bolts securing the bracket to the cabinet.
6. Remove the three bolts securing the removable skid and remove the skid.

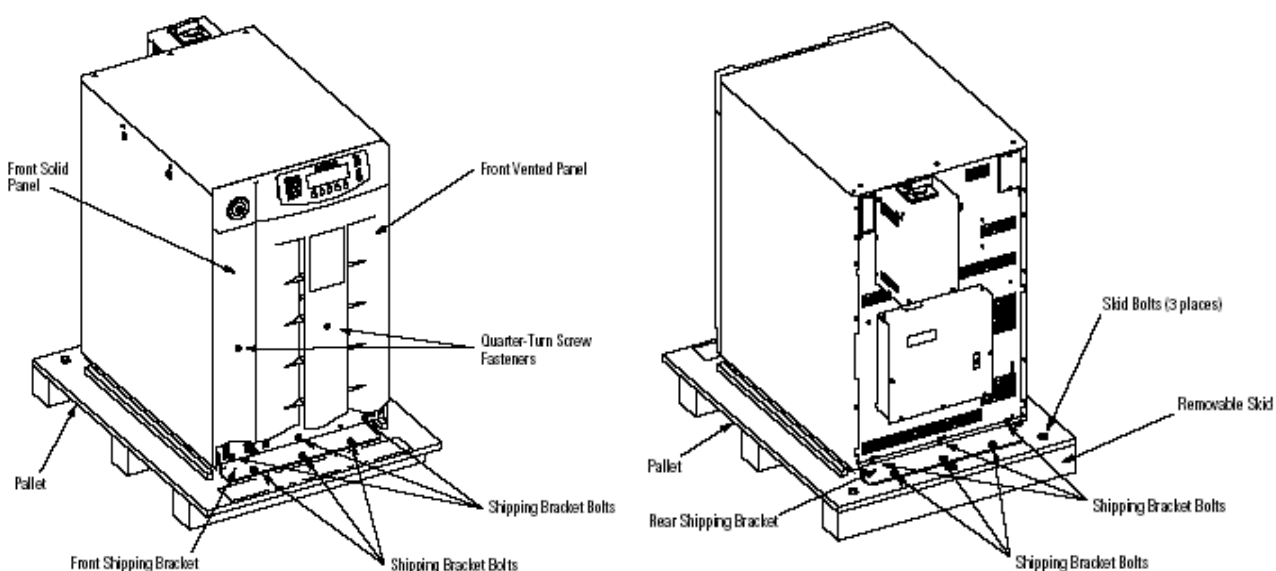


Figure 2 - UPS Shipping Bracket Removal

NOTE *In the following steps the pallet tilts and acts as a ramp once the UPS is rolled beyond the center of the pallet. Be sure to restrain the UPS as it continues to roll down the pallet / ramp. The front shipping bracket acts as a brake to assist restraining the UPS.*

WARNING

Do not stand directly behind the pallet while unloading the cabinet. If unloading instructions are not closely followed, the UPS may roll unexpectedly and cause serious injury.

7. Slowly roll the UPS toward the rear of the pallet. Once the pallet tilts, continue rolling the UPS down the ramp until it clears the pallet.
8. Remove the three bolts securing the front shipping bracket to the cabinet.
9. If installing the UPS permanently, retain the hardware for later use; otherwise, discard or recycle the pallet and hardware in a responsible manner.
10. Roll the UPS to its final installation location.
11. Secure it in position by lowering the leveling feet until the UPS is not resting on the casters and it is level.
12. If a more secure installation is desired, the retained hardware may be used to secure the UPS cabinet to the floor.
13. Reinstall the front and rear shipping brackets to the UPS with the angles facing outward.
14. Attach the angle brackets to the floor with contractor-supplied hardware.

1.4 Connecting UPS Internal Batteries

The UPS ships from the factory with the batteries disconnected internally. These must first be connected before the UPS can be used.

Note: The following information has been extracted from the Powerware “Installation and Operation Manual” provided with the UPS. Refer to Section 3.5 of that document, for additional information if necessary.

To connect the battery strings:

1. Verify that all battery breakers are in the OFF position.
2. If not already removed, remove the left front solid panel and the right front vented panel from the UPS cabinet.
3. Front panels are secured with quarter-turn screw fasteners and magnetic latches. To remove the panels, unscrew the fasteners counterclockwise and pull the panels straight forward to disengage the magnetic latches.
4. Remove the screws securing the internal metal safety shield and remove it to gain access to the batteries. Retain the hardware for later use.

Note: Some early production units have a one-piece metal safety shield covering the circuit breakers and battery compartment. The newer two-piece configuration is shown below.

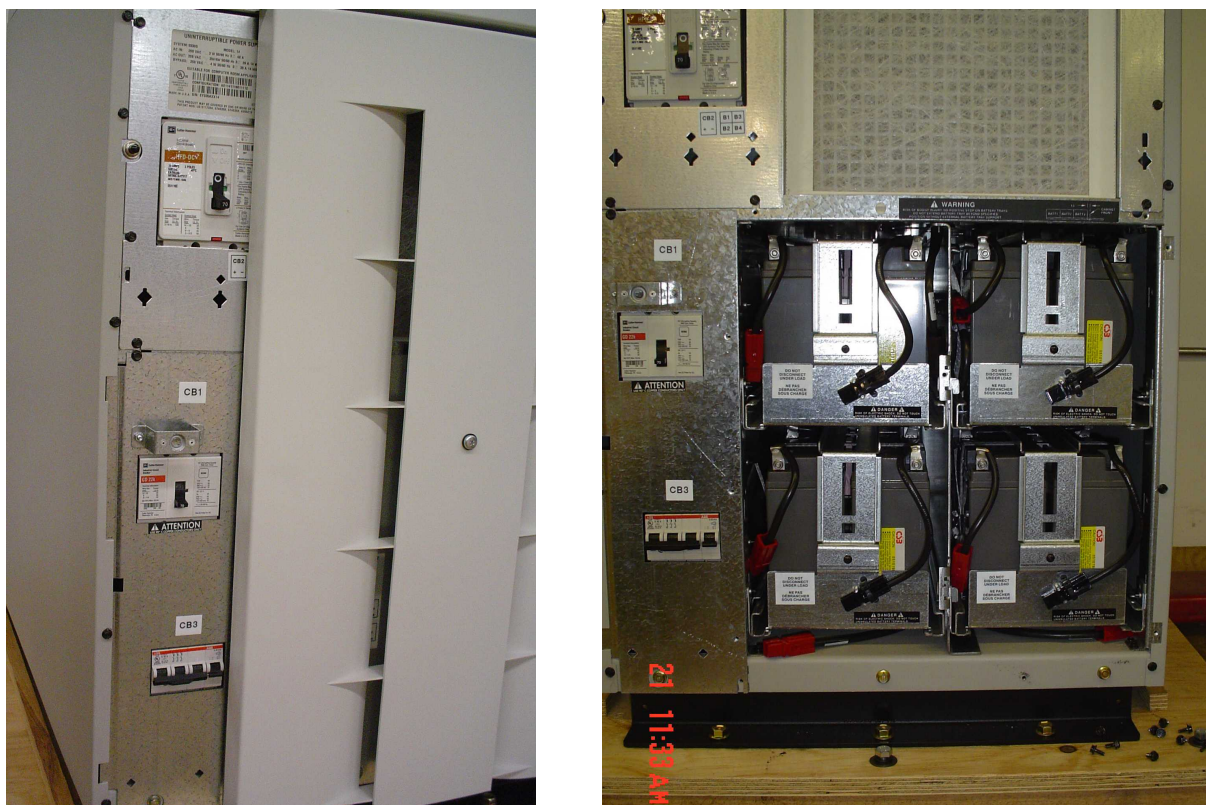


Figure 3 - UPS Front Cover Removal & Battery Compartment

5. Connect the string of battery trays together by mating the loose red connector from each tray to the fixed black connector mounted on the front edge of the tray above it.

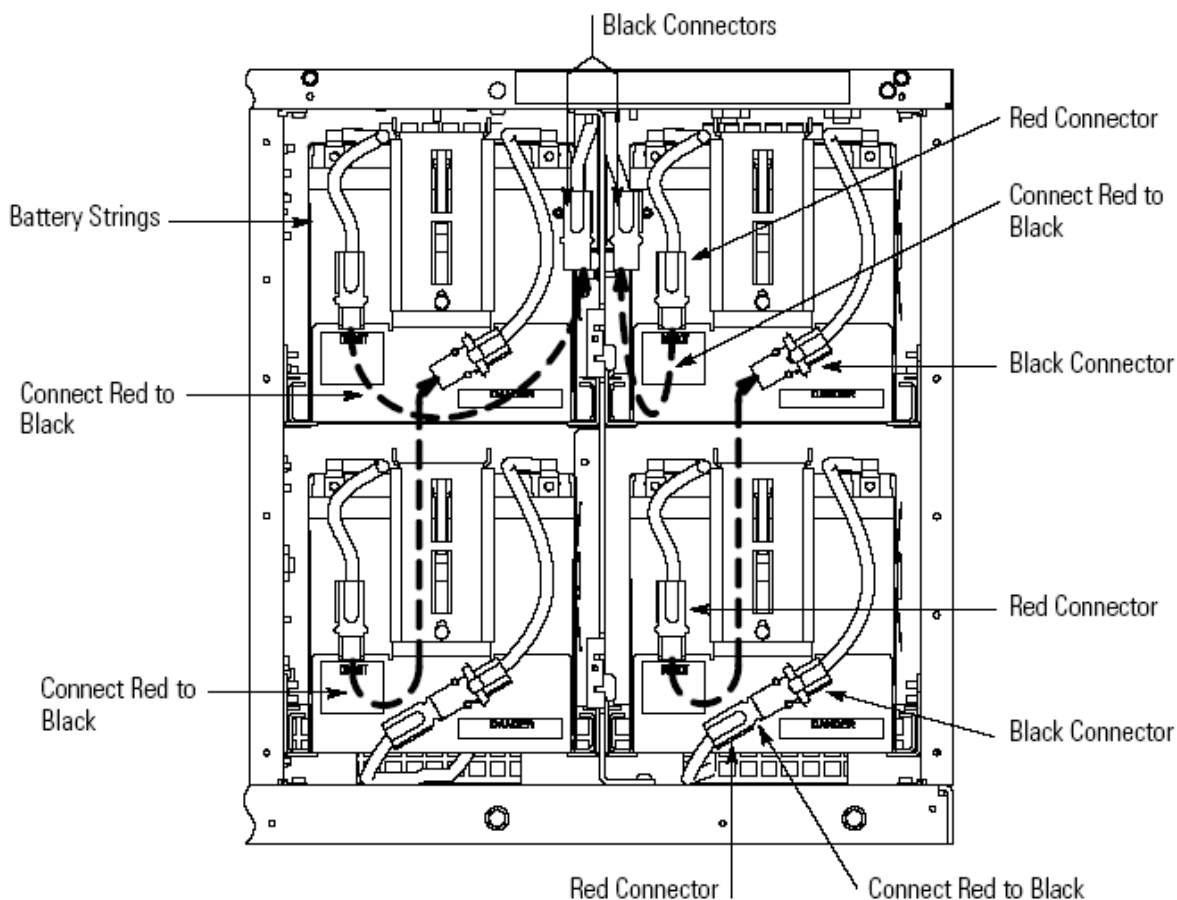
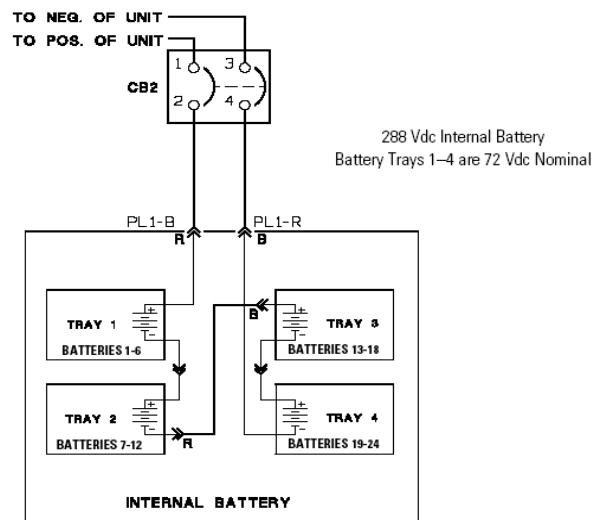


Figure 4 - UPS Battery Connections

Figure 5 - UPS Battery Schematic



6. Reinstall the internal safety shields and right front panel.
Leave the smaller, left front cover off until after start-up is complete.

Section 2 - Powerware 9330G-14B UPS Connections

2.1 UPS I/O Terminal Compartment

All power and control connections are made in the I/O terminal compartment located on the rear of the UPS. To access the compartment, remove the seven machine screws holding the cover in place and set aside for later use.

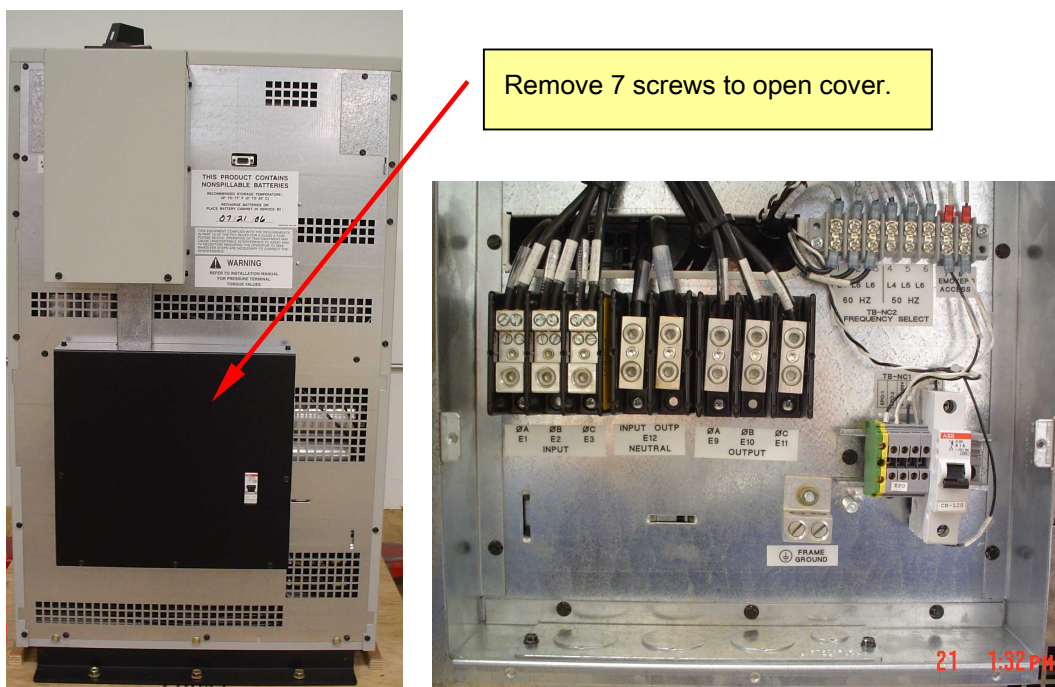


Figure 6 - UPS I/O Terminal Compartment

2.2 UPS Harmonic Filter Connection

The 9330G-14B UPS is equipped with an input harmonic filter. To optimize overall performance, this filter must be connected to match the mains frequency, 50 or 60Hz. When shipped from the factory it is normally connected for 60Hz operation. Verify the Filter Frequency Selection connections shown below match the mains frequency of the site and adjust as required.

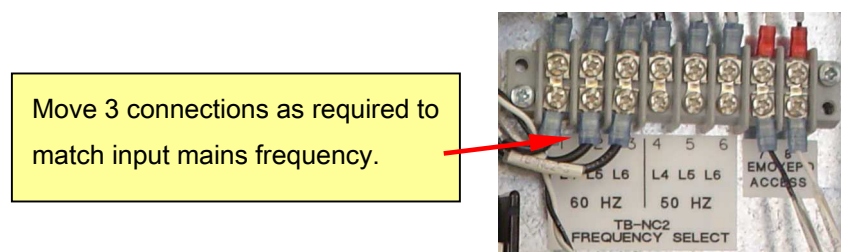


Figure 7 - UPS Harmonic Filter Connections

2.3 UPS Input Power Cable (from PDU)

Connect UPS Input Power Cable (P/N 5125079) as follows:

- 1) Pull cable end labeled "TO UPS INPUT" through the left most 3/4" cord grip in the back of the UPS.
- 2) Connect the BLACK wire to INPUT A.
- 3) Connect the RED wire to INPUT B.
- 4) Connect the ORANGE wire to INPUT C.
- 5) Connect the BLUE wire to INPUT NEUTRAL.
- 6) Connect the GREEN/YELLOW wire to the left GND lug.
- 7) Securely tighten the cord grip on the input cable.

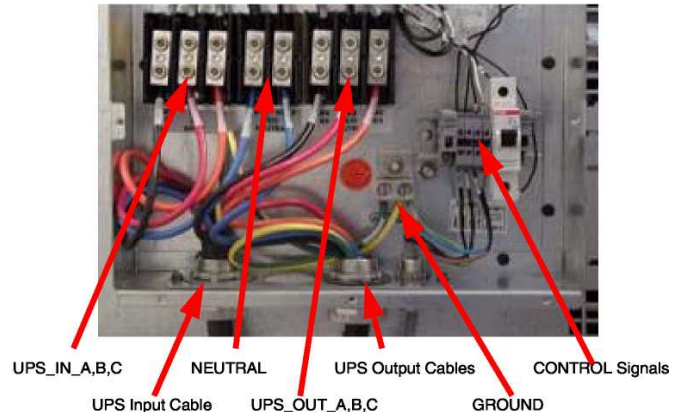


Figure 8 - UPS Power Connections

2.4 UPS Output Power Cable (to PDU)

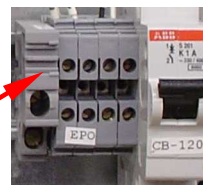
Connect UPS Output Power Cable (P/N 5125079-2) as follows:

- 1) Pull cable end labeled "To UPS OUTPUT" through the right 3/4" cord grip in the back of the UPS.
- 2) Connect the BLACK wire to OUTPUT A.
- 3) Connect the RED wire to OUTPUT B.
- 4) Connect the ORANGE wire to OUTPUT C.
- 5) Connect the BLUE wire to OUTPUT NEUTRAL
- 6) Connect the GREEN/YELLOW wire to the right GND lug.
- 7) Securely tighten the cord grip on the input cable.

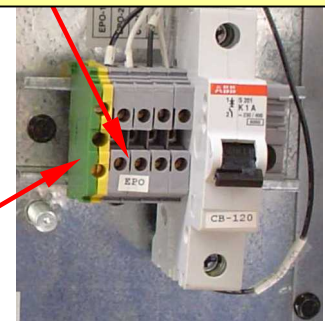
2.5 UPS Control Cable to A1 Main Disconnect & UPS Control Panel

Connect UPS Control cable (P/N 5169224) as follows:

- 1) Route the cable end labeled "To UPS" through the 1/2" cord grip in the bottom of the UPS Terminal Box and route to control terminal strip, TB-NC1
- 2) Connect the GREEN/YELLOW (GND) and SHIELD wires to the Grounding Terminal.
- 3) Remove jumper and connect the BLK 1 (EPO_1) wire to EPO-1.
- 4) Remove jumper and connect the BLK 2 (EPO_2) wire to EPO-2.
- 5) Connect the BRN (AUX 120) wire to CTRL-120V.
- 6) Connect the BLU (NEUT) wire to CTRL-RTN.
- 7) Securely tighten the cord grip onto the cable.



GND, EPO-1, EPO-2, 120V, NEUT.

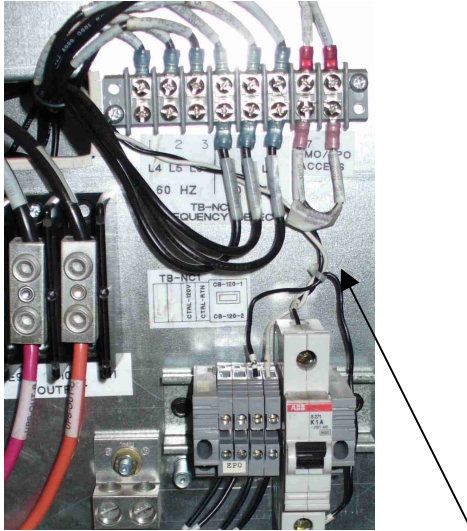


NOTE: If not previously installed, replace DIN Rail Terminator with Ground Block provided in kit, as required.

Figure 9 - UPS Control Connections

When all connections are secure, re-install the terminal box cover at the rear of the UPS.

2.6 UPS Control Cable to A1 Main Disconnect & Jumper Removal



- 1) Remove this EPO Jumper if present
- 2) Confirm that both EPO jumpers were removed.

Warning

Failure to remove both jumpers will prevent the UPS from disconnecting during system power off.

Chapter 3

NGPDU - Power Distribution Unit

This UPS kit has been specifically designed for compatibility with DST or LightSpeed 7.x (VCT, Pro³² & DVCT) series CT & PET/CT systems. These systems utilize an NGPDU. Model numbers currently in use are:

- 2326492-3
- 2326492-4
- 2326492-5

The same interconnection information applies to all listed NGPDU models.

Section 1 - PDU Connections

If not done previously, shutdown system software and turn off the mains input power to the system.

NOTE: Use proper Tag and Lock-Out safety procedures!

1.1 UPS Input Power Cable

- 1) Remove the top and front covers of the PDU.
- 2) Remove all three jumper wires at terminal strip TS4, terminals 1 to 6, 2 to 7, & 3 to 8. Discard the jumpers.
- 3) Route the UPS Input Power Cable (P/N 5125079) into the PDU and up through the indicated cable clamp in the bottom pan. Refer to the Figure below.
- 4) Connect the BLACK wire to TS4-1.
- 5) Connect the RED wire to TS4-2.
- 6) Connect the ORANGE wire to TS4-3.
- 7) Connect the BLUE wire to TS4-4.
- 8) Connect the GREEN/YELLOW wire to TS4-5.

UPS Input Power Cable (PDU -> UPS)

UPS Output Power Cable (UPS -> PDU)

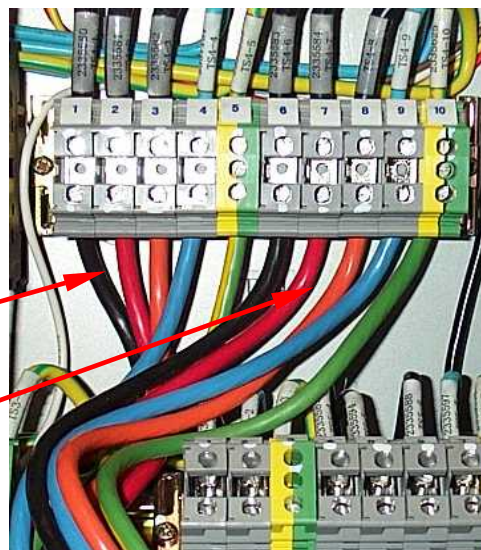


Figure 10 - NGPDU Connections

1.2 UPS Output Power Cable

- 1) Route the UPS Output Power Cable (P/N 5125079-2) into the PDU and up through the indicated cable clamp in the bottom pan.
- 2) Connect the BLACK wire to TS4-6.
- 3) Connect the RED wire to TS4-7.
- 4) Connect the ORANGE wire to TS4-8.
- 5) Connect the BLUE wire to TS4-9.
- 6) Connect the GREEN/YELLOW wire to TS4-10.

Note: Do not re-install the PDU covers at this time. Wait until after the system start-up procedures have been completed.

Chapter 4

A1 Disconnect Panel

Section 1 - A1 Main Disconnect Panel

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Main Disconnect Panel. Use of one of the following GE panels, or equal, specifically designed for use with this kit, is recommended. Refer to your system Pre-Installation manual for proper selection.

- E4502AB / P5051RJ (90A)
- E4502AC / P5051RK (110A)
- E4502AE (125A)
- E4502AF (150A)



Figure 11 - Typical A1 Panel Front View & Interior View.

DANGER

THE FOLLOWING PROCEDURES ARE ACCOMPLISHED INSIDE THE A1 DISCONNECT PANEL. MAINS VOLTAGE MAY STILL BE PRESENT EVEN WITH THE MANUAL SWITCH PULLED. LOCATE THE FEEDER DISCONNECT TO THE A1 PANEL AND SHUT OFF MAIN POWER THERE. USE PROPER LOCK-OUT / TAG-OUT PROCEDURES. TEST THE A1 DISCONNECT TO VERIFY THAT ALL POWER IS OFF BEFORE PROCEEDING.

1.1 UPS Control Cable Connections to A1 Panel

An electrician will be required to install a 3/4" conduit or raceway between the A1 Disconnect and the floor in the general area of the UPS. Conduit should be terminated in a small wall box. This route will be used for the UPS Control cable connecting the UPS and A1 Main Disconnect Panel.

WARNING

REMOVE THE FACTORY INSTALLED JUMPER BETWEEN TERMINALS TB1-7 & TB1-9 IN THE A1 PANEL, IF PRESENT. THIS IS NECESSARY TO PREVENT DAMAGE TO THE UPS EPO CIRCUITRY.

- 1) Install a 1/2" cord grip (supplied with the kit) in the wall box or the base of the A1 Panel.
- 2) Route the UPS Control Cable (P/N 5169224) from the UPS Disconnect Panel, through the conduit or raceway to the A1 Disconnect Panel.
- 3) Connect the BLK 1 wire to terminal TB1-7 in the A1 Panel.
- 4) Connect the BLK 2 wire to terminal TB1-8 in the A1 Panel.
- 5) Connect the BRN wire to terminal TB1-9 in the A1 Panel.
- 6) Connect the BLU wire to terminal TB1-10 in the A1 Panel.
- 7) Connect the GREEN/YELLOW (GND) and SHIELD wires to terminal TB1-11 (GND) in the A1 Panel.

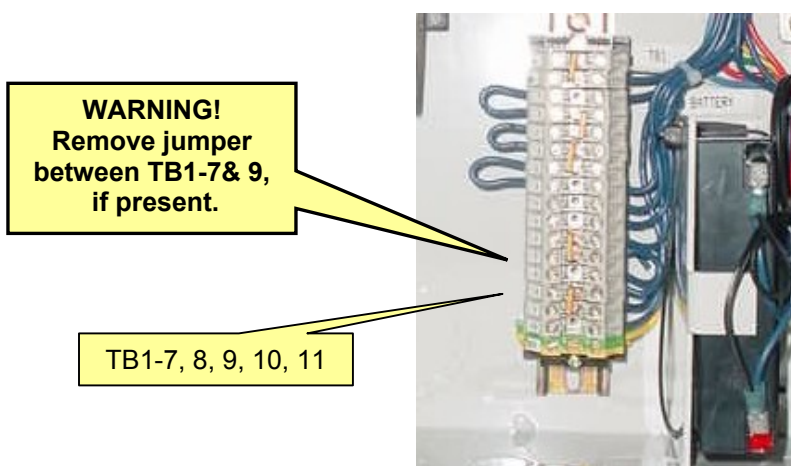


Figure 12 - A1 Panel TB1 Terminals

1.2 A1 Panel Voltage Selection

Verify the A1 Panel Control Transformer is configured for the correct nominal mains voltage. The A1 panel ships from the factory configured for 480V input. Reconnect for 400V if required.

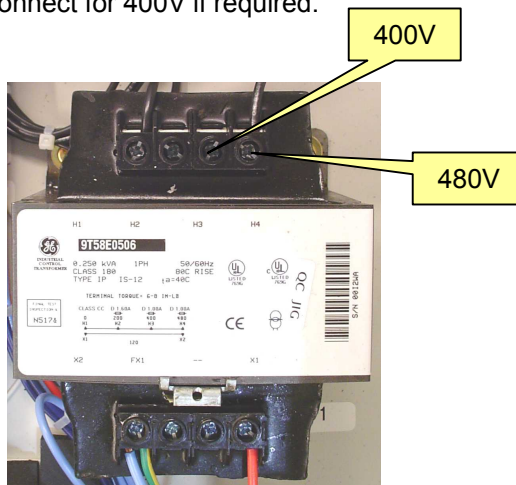


Figure 13 - A1 Panel Control Transformer

1.3 A1 Panel Labeling

A pre-printed adhesive label is provided in the UPS kit titled “SYSTEM OFF PROCEDURE” and “SYSTEM START-UP PROCEDURE”. Attach the label to the front of the A1 Panel as indicated below.

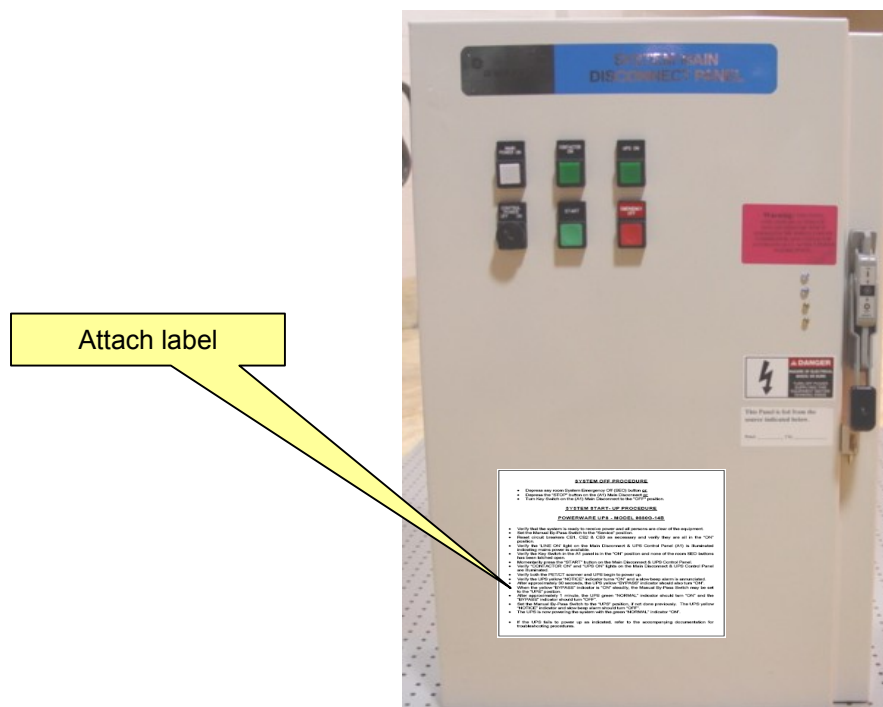


Figure 14 - A1 Panel Labeling

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Chapter 5

Start Up / Shutdown Procedures

NOTE:

GEHC installation team and service employees are authorized to perform initial power-up of the Powerware 9330G-14B UPS. It is not necessary to call a Powerware representative for initial start-up. However, if you prefer or have difficulty with the UPS start-up procedure, you may arrange for a Powerware service technician to perform start-up, at no cost. The Powerware contact number in the US is 1-800-843-9433. For international service refer to the contact information provided with the Powerware documentation.

Section 1 - Start-Up: Powerware 9330G-14B

1.1 Automatic Configuration Mode

Automatic configuration is performed during UPS startup. The UPS will automatically sense the input voltage and frequency and match its output to the mains input power.

No additional frequency or voltage configuration set-up is required.

1.2 Auto Re-start

The UPS has been configured at the factory to automatically start-up whenever it senses input power. If the UPS does not automatically start, Powerware service should be contacted to correct the problem and verify proper operation.

1.3 Initial Start-up Sequence

The following procedure should be used to power up the system up for the first time after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off. Turn OFF the Console power switch. This keeps the system computers from needlessly booting up until proper UPS system operation is verified.

Note: It is strongly recommended to have the "Installation and Operation Manual" for the UPS available.

Check the following before proceeding to power up the system:

- Mains input power was turned OFF during installation. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the A1 connections procedure.
- 1) Verify that the system is ready to receive power and all persons are clear of the machine.
- 2) Set the UPS Manual By-Pass Switch to the "Service" position.
- 3) Verify the Auxiliary circuit breaker at the back of the UPS is in the "ON" position.
- 4) Reset UPS circuit breakers CB1, CB2 & CB3 as necessary and verify they are all in the "ON" position.
- 5) Verify the 'LINE ON' light on the Main Disconnect & UPS Control Panel (A1) is illuminated indicating mains power is available.
- 6) Verify the Key Switch in the A1 panel is in the "ON" position and none of the room SEO buttons has been latched open.
- 7) Momentarily press the "START" button on the Main Disconnect & UPS Control Panel.

- 8) Verify 'CONTACTOR ON' light on the A1 Disconnect & UPS Control Panel is illuminated.
- 9) Verify both the UPS and CT or PET/CT scanner begins to power up. If the UPS "ALARM" indicator stays on or the UPS fails to start up properly, check for an **"Input Phase Rotation"** or other alarm message as described below.
- 10) Verify the UPS yellow "NOTICE" indicator turns "ON" and a slow beep alarm is annunciated.
- 11) After approximately 30 seconds, the UPS yellow "BYPASS" indicator and A1 Panel "UPS ON" indicator should turn "ON".
- 12) When the yellow Bypass indicator is ON steadily, the Manual By-Pass Switch may be set to the "UPS" position.
- 13) After approximately 1 minute, the UPS green "NORMAL" indicator should turn "ON" and the "BYPASS" indicator should turn "OFF".
- 14) Set the Manual By-Pass Switch to the "UPS" position, if not done previously. The UPS yellow "NOTICE" indicator and slow beep alarm should turn "OFF".
- 15) The UPS is now powering the system with the green "NORMAL" indicator "ON".

NOTE - The UPS is now providing 120VAC power to the Table, Gantry and Console. In the event of a momentary mains power loss, scanning will be inhibited or aborted. However, the UPS will continue to provide power to the Table, Gantry & Console for at least 10 minutes. The UPS will "beep" occasionally, letting you know that it is on BATTERY BACKUP. If mains power is restored prior to shutting down the CT or PET/CT system, the UPS and system will automatically return to normal operation.

1.4 UPS Alarm Conditions

1.4.1 Input Phase Rotation

If the UPS does not start properly and the display indicates an **"Input Phase Rotation"** message, there is an input phase error.

Two of the main input phases must be interchanged at the **PDU MAINS INPUT** Terminal Block. Do **NOT** reverse any of the connections at the UPS or PDU TS4 Interface.

1.4.2 Other Alarms

Refer to the 9330G-14B "Installation and Operation Manual" for a detailed listing of operational and diagnostic information for the UPS.

Section 2 - Shut-Down Procedure

2.1 Normal System Shutdown

To shut down the entire system, do the following:

- 1) Perform a software shutdown.
- 2) Turn off the Console power switch.
- 3) Press any room System Emergency Off (SEO) button or
- 4) Press the STOP button on the A1 Main Disconnect Panel or
- 5) Set the manual switch of the A1 Disconnect Panel in the OFF position.

To re-start the system, follow the Start-Up instructions in the previous section.

2.2 UPS Shutdown Only – Manual Bypass

To shut down the UPS only while leaving the system operational, do the following:

- 1) Turn the Manual Bypass switch to the BYPASS position.
- 2) Turn OFF all UPS circuit breakers.

Comment:

Although the UPS stops supplying power and is disconnected internally from the batteries, system power still passes through the UPS cables while in bypass mode. Use caution when servicing.

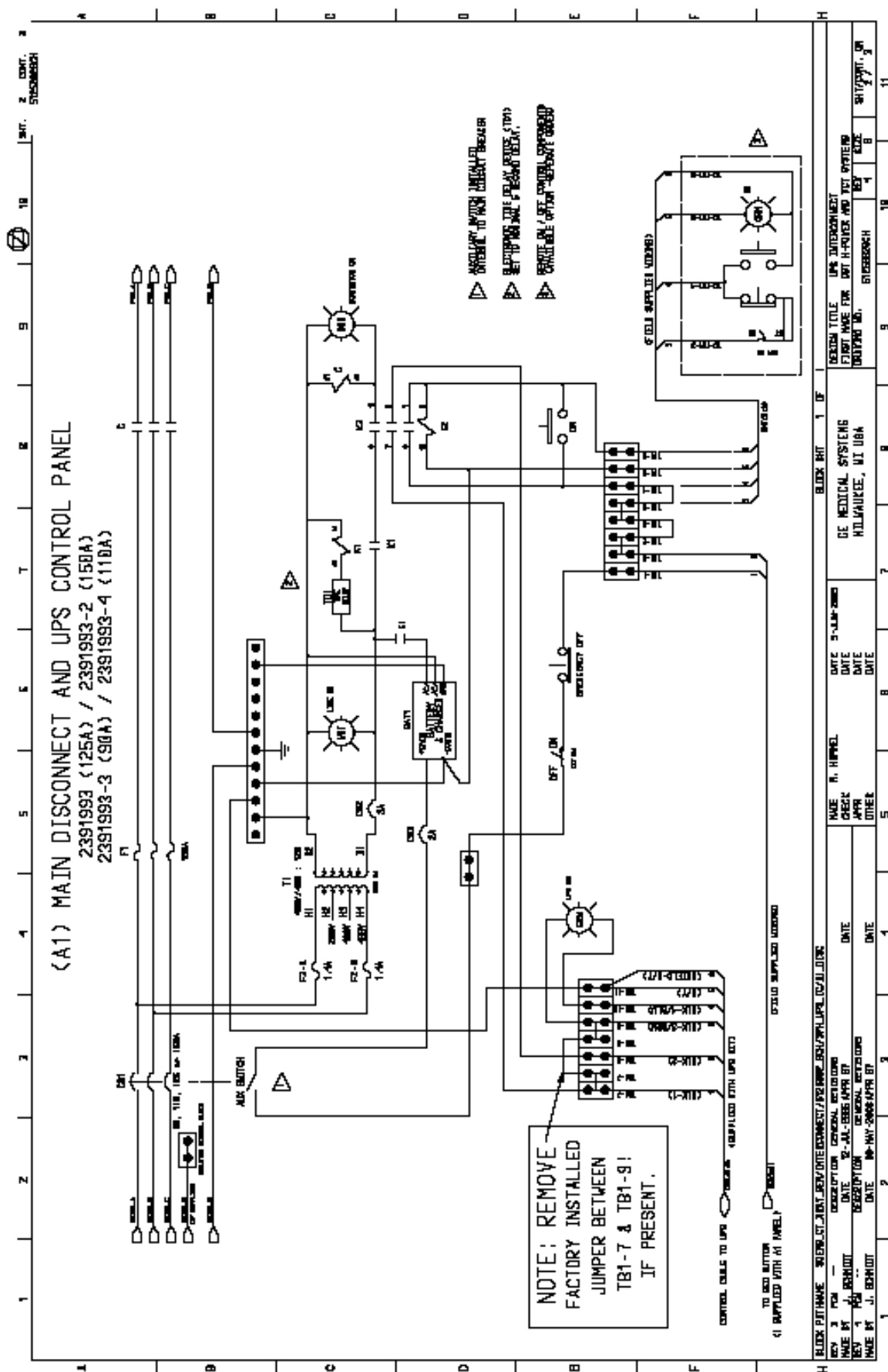
2.3 Final Steps

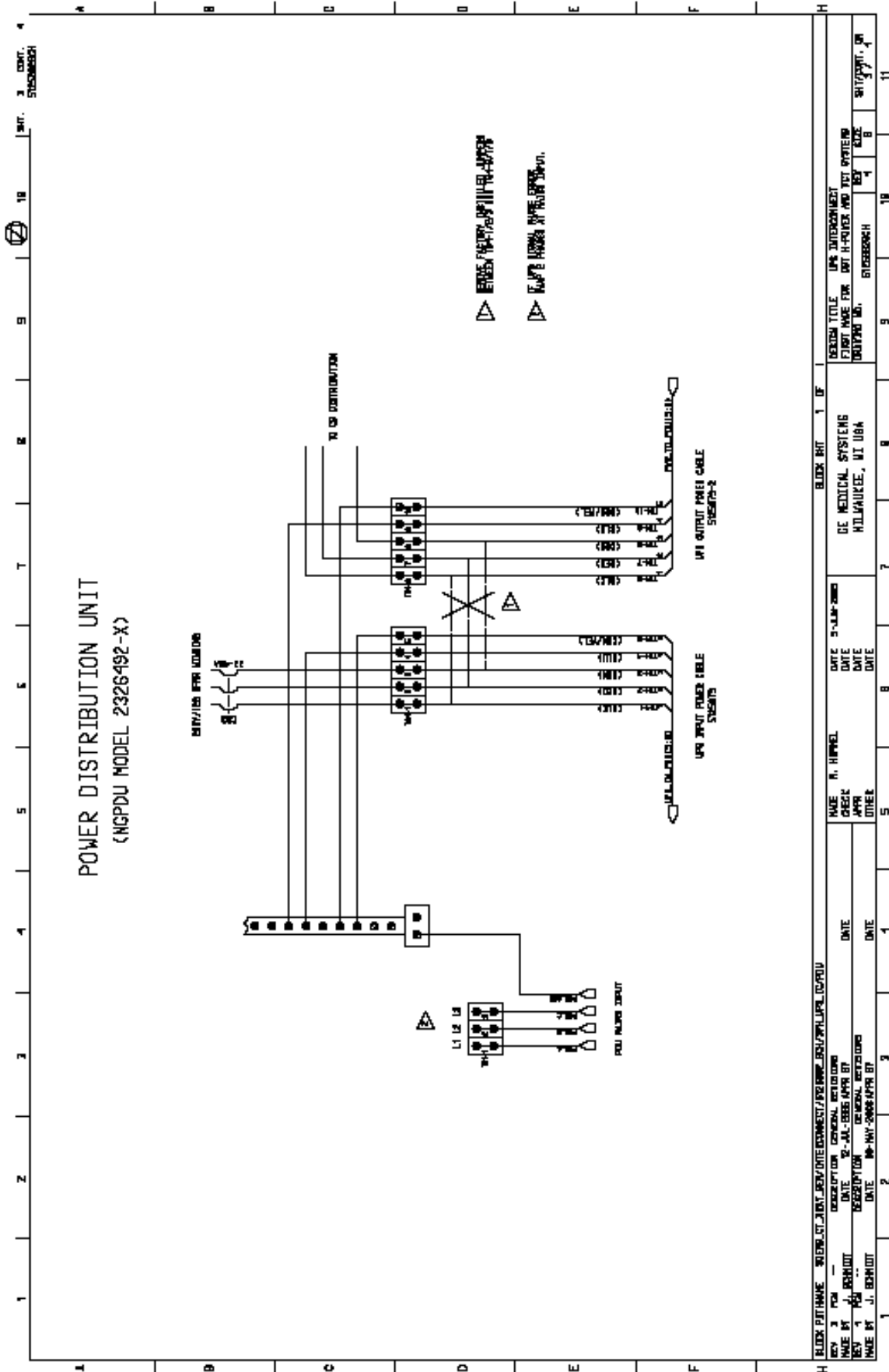
- 1) Re-install all safety shields and covers on the UPS and PDU.
- 2) Fill out and mail the Powerware Warranty Registration card or register the UPS on-line at www.warranty.powerware.com.

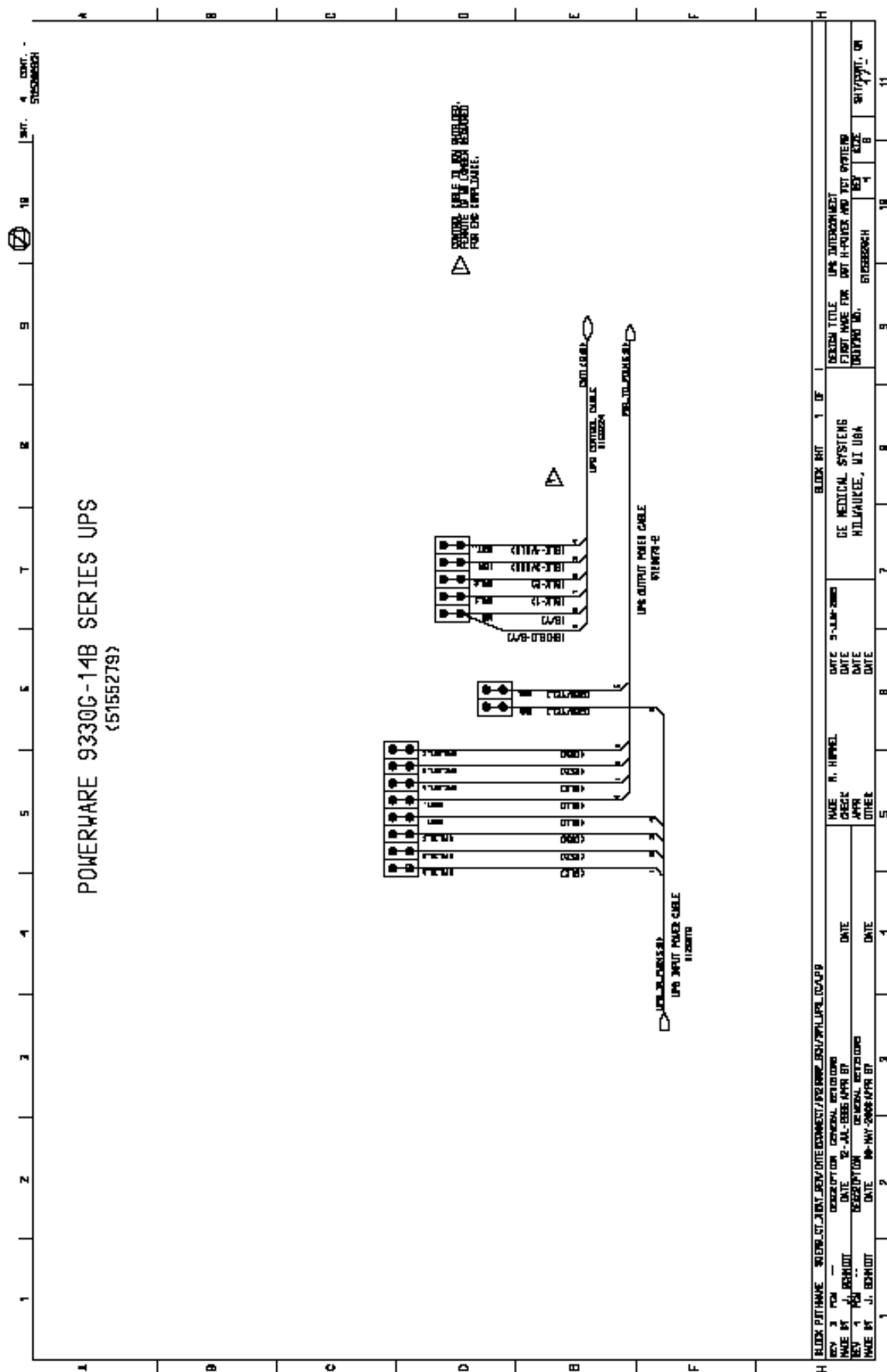
Chapter 6

Schematic 5125082SCH, rev 4

Figure 15 - Schematic 5125082SCH, rev 4 begins on the following page.







Appendix A

Applicable Documents

The following documents provide reference information for the UPS Interconnect Kit.

Section 1

GEHC Reference Documents

- 5125082SCH UPS Interconnect Schematic
- 5155279PSP 3Ph, 14kVa UPS Purchase Specification
- 2326492-xSCH NGPDU Schematic(s)
- 2391993PSP Purchase Specification for A1 Main Disconnect & UPS Control Panel
- 5125083ADW Assembly Specification for the UPS Hardware Kit
- 46-015900 Global Documentation Standard
- 46-131001 Applied Practices (Assemblies)
- 46-170700 Applied Practices (Portions)

Section 2

Vendor Documents

- POWERWARE 9330G-14B Installation and Operation Manual

Section 3

Industry Standards

- NFPA-70 (2005) National Electrical Code (USA)

Appendix B

Field Replaceable Unit (FRU) Listing

The following listing identifies the Field Replaceable Units (FRUs) associated with this assembly. Vendor part numbers are included as examples only. For the complete list of approved vendors for each item, see the associated Manufacturer's Equivalent on GEHC ePDM system.

GEMS PART #	NAME / DESCRIPTION	VENDOR	VENDOR P / N
5155279	PW9330G-14B UPS	POWERWARE	AE1141116011112
5125083	Hardware Kit	GE Supply	5125083

Figure 16 - Field Replaceable Units

GE Medical Systems, a General Electric Company, going to market as GE Healthcare.
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com

